# Semantic Transfer Learning

Gary Nan Tie, Jan 7, 2026

**Abstract**
For the purposes of zero-shot next-token and sequence-to-sequence learning, Noken implicit deep learning models, can be composed and connected into feedback loops, enabling a myriad of network configurations to be explored, and so find ways to lower average entropy. Beyond statistical correlations, this novel attention mechanism is based on semantic architecture.

Suppose we have learned the semantics of two different subject domains and want to explore interdisciplinary inquiries. Without starting over from scratch and training on the all combined data (which may in fact muddle matters) we introduce an algorithm to leverage and transfer the individual semantics to make sense of the interdisciplinary corpus.

# Introduction

Generic Implicit deep learning model:
for a given input vector u,
$x = \varphi(Ax+Bu)$ [equilibrium equation]
$y(u) = Cx+Du$ [prediction equation]
where $\varphi : R^n \to R^n$ is a nonlinear vector map (the "activation" map), assumed to be blockwise Lipschitz, and matrices A,B,C,D contain model parameters to be learned, A assumed to be well-posed for $\Phi$, so that the equilibrium equilibrium has a unique solution.

Let u = ti be a token from context C = {t1, t2, … , tm}
and y = ε, an n-vector of positive reals, in an implicit model:
$si = \Phi(Vsi + Qti)$
$\varepsilon = K'si + Kti$
Let query qi = Qti, key ki = -Kti, creating pairs (qi, ki),
and let value vi = Vsi, key k'i = K'si, creating pairs (vi, k'i),
the respective keys providing semantics for queries and values.
We call the latent token si corresponding to ti, a Noken.

Given i, let αij = softmax score(qi, k'j), where score is a similarity function.
Let hi = Sum αij vj and βij = softmax score(hi, kj).
Let average Renyi α-entropy of {ti} be:
E{ti} = E(βij) = (1/m) Sum H(I, α), where H(I, α) = Hα(βij),
Hα is Renyi α-entropy.

These Noken implicit models can be composed and connected into feedback loops, enabling a myriad of network configurations to explore and find ways to lower entropy.

Suppose Alice trained her Noken semantics on context CA and Bob his on CB. Now consider context CAB a subset of CA U CB, rather than starting over and learning a Noken implicit model on CAB, can we leverage what Alice and Bob have already learned on CA and CB respectively, and transfer their semantic learning to CAB? As we shall see, the short answer is yes, provided Alice and Bob are both happy with what they've learned.

Alice trains an implicit deep learning model

on context $C_A = \{ t_1^A, \ldots, t_{m_A}^A \} \subseteq \mathbb{R}^r$

Let $n > r$ and $\varepsilon_A = \begin{bmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_n \end{bmatrix} \in \mathbb{R}^n_+$

She learns matrices $\underset{n\times n}{V_A}$, $\underset{n\times n}{\Lambda_A}$, $\underset{n\times r}{Q_A}$, $\underset{n\times r}{K_A}$,

and fixed point state $\underset{n\times 1}{s_i^A}$ so that for $i \in [m_A]$:

$$s_i^A = \varphi_A ( V_A s_i^A + Q_A t_i^A )$$

$$\varepsilon_A = \Lambda_A s_i^A + K_A t_i^A$$

where $V_A$ is well-posed for activation function $\varphi_A$.

Let query $q_i^A = Q_A t_i^A$ and key $k_i^A = -K_A t_i^A$,

creating pairs $(q_i^A, k_i^A)$.

Let value $v_i^A = V_A s_i^A$ and key $\lambda_i^A = \Lambda_A s_i^A$,

creating pairs $(v_i^A, \lambda_i^A)$.

The keys $k^A$ provide semantics for queries $q^A$ and

the keys $\lambda^A$ provide semantics for values $v^A$.

We call the latent token $s_i^A$, corresponding to

token $t_i^A$, a <u>Noken</u>.

Similarly, Bob has an implicit model for $i \in [m_B]$:

$$s_i^B = \varphi_B \left( V_B s_i^B + Q_B t_i^B \right)$$

$$\varepsilon_B = \Lambda_B s_i^B + K_B t_i^B$$

where $V_B$ is well-posed for activation $\varphi_B$.

## Semantic Attention:

Given $i \in [m_A]$, let $\alpha^A_{ij} = \underset{j \in [m_A]}{\text{softmax}}\ \text{score}(q^A_i, k^A_j)$

where score is a similarity function, like a kernel.

The contextual representation of token $t^A_i$ is:

$$h^A_i \triangleq \sum_{j=1}^{m_A} \alpha^A_{ij} v^A_j$$

Let $\beta^A_{ij} = \underset{j \in [m_A]}{\text{softmax}}\ \text{score}(h^A_i, k^A_j)$

and $j^{\beta^A}_i = \underset{j \in [m_A]}{\arg\max}\ \beta^A_{ij}$.

( Similarly for Bob, define $\beta^B_{\cdot}$ )

Let the average Renyi $\alpha$-entropy of $\{t^A_i\}$ be:

$$E\{t^A_i\} = E(\{\beta^A_{ij}\}) = \frac{1}{m_A} \sum_{i=1}^{m_A} H^i_\alpha$$

where $H^i_\alpha = H_\alpha(\{\beta^A_{ij}\}_{j=1}^{m_A})$

and $H_\alpha$ is Renyi $\alpha$-entropy.

Interdisciplinary Attention:

Let $t_p \in \{t_1, \ldots, t_{m_{AB}}\} =: C_{AB} \subseteq C_A \cup C_B$

i) If $t_p = t_i^A \in C_A \setminus C_B$

define output token $o_{p+1} = t^A_{j_i^{\beta A}}$

ii) If $t_p = t_k^B \in C_B \setminus C_A$

define output token $o_{p+1} = t^B_{j_k^{\beta B}}$

iii) If $t_i^A = t_p = t_k^B$ ie $t_p \in C_A \cap C_B$

let $\rho_A = \frac{m_A}{m_A + m_B} \beta^A_{i\, j_i^{\beta A}}$ and $\rho_B = \frac{m_B}{m_A + m_B} \beta^B_{k\, j_k^{\beta B}}$

to account for sample size.

case: $\rho_A > \rho_B$, $o_{p+1} \triangleq t^A_{j_i^{\beta A}}$

case: $\rho_B > \rho_A$, $o_{p+1} \triangleq t^B_{j_k^{\beta B}}$

case: $\rho_A = \rho_B$, Flip a fair coin, $o_{p+1} = \begin{cases} t^A_{j_i^{\beta A}} & \text{heads} \\ t^B_{j_k^{\beta B}} & \text{tails} \end{cases}$

Input token $t_p \mapsto$ Semantic Attention $\longmapsto$ Output token $o_{p+1}$
(on $C_{AB}$)

The attention mechanism on $C_{AB}$ just described is impartial to Alice and Bob and optimally aligns their semantics. If $E\{t_i^A\}$ and $E\{t_k^B\}$ are both acceptable to Alice and Bob then they would agree to this semantic transfer learning on $C_{AB}$.

On the other hand, one could start over on data $C_{AB}$ and train another implicit model, at additional computational expense. An upside is that semantic interactions between $C_A$ and $C_B$ may emerge. A downside is that semantic clarity of either $C_A$ or $C_B$ may be muddied by spurious relationships.

Nonetheless, semantic transfer learning via attention on $C_{AB}$ above, contributes to Noken interlingua.

$\square$

## References

[1] 'Semantic Attention via Noken Nen'
Gary Nan Tie, Dec 23,2025
DOI: 10.13140/RG.2.2.10508.99202
Noken, implicit deep learning models, can be composed into cascades, as well as connected into feedback loops, like Zen Nen. Given a target input token from a context string, we can take a draw from a distribution over the string centered at the target, to provide auxiliary contextual information. A Noken, a latent token representing the target, can be potentially refined in a cascade, or iterated in a feedback loop, to utilize the auxiliary information of the drawn token. Convolution like network configurations are explored to reduce average Renyi α-entropy, for the purposes of zero-shot next-token and sequence-to-sequence learning. Beyond statistical correlation, Noken networks introduce a novel attention mechanism based on semantic architecture.

[2] 'Noken squared:
Implicit implicit contextual token learning'
Gary Nan Tie, Dec 10, 2024
DOI: 10.13140/RG.2.2.20497.49768/1
'Noken thinking machines'
DOI: 10.13140/RG.2.2.10400.85765/1
Composing implicit token models is a way to update queries in a query-key-value attention mechanism. Coherent refinement is achieved when Renyi α-entropy is reduced. This approach jointly learns query, key, value embeddings and query updates, coordinated by concatenated implicit models.

[3] 'Noken: implicit contextual token learning'
Gary Nan Tie, Dec 4, 2025
DOI: 10.13140/RG.2.2.17440.24329/1
'Noken variations'
DOI: 10.13140/RG.2.2.33204.92805/1
Given a token in a context string, an implicit deep learning model reveals a latent token, that we dub a 'noken', and token representations, such that keys embue meaning to queries and values. This construction can then be construed as a machine oracle that aligns meaning utilizing a universal interlingua for next-token and sequence-to-sequence learning.